To compare the dispersion of two data, coefficient of variation $= \frac{\sigma}{\bar{x}} \times 100\%$

The coefficient of variation of $A = \frac{191.5}{700} \times 100\% = 27.4\%$

The coefficient of variation of $B = \frac{114.9}{420} \times 100\% = 27.4\%$

Thus the two data have equal coefficient of variation. Since the data have equal coefficient of variation values, we can conclude that one data depends on the other. But the data values of B are exactly 60% of the corresponding data values of A. So they are very much related. Thus, we get a confusing situation.

To get clear picture of the given data, we can find their coefficient of variation. This is why we need coefficient of variation.

Progress Check

1. Coefficient of variation is a relative measure of _____.
2. When the standard deviation is divided by the mean we get _____.
3. The coefficient of variation depends upon _____ and _____.
4. If the mean and standard deviation of a data are 8 and 2 respectively then the coefficient of variation is _____.
5. When comparing two data, the data with _____ coefficient of variation is inconsistent.

Example 8.15 The mean of a data is 25.6 and its coefficient of variation is 18.75. Find the standard deviation.

Solution Mean $\bar{x} = 25.6$, Coefficient of variation, C.V. = 18.75

$$\text{Coefficient of variation, C.V.} = \frac{\sigma}{\bar{x}} \times 100\%$$

$$18.75 = \frac{\sigma}{25.6} \times 100 \Rightarrow \sigma = 4.8$$

Example 8.16 The following table gives the values of mean and variance of heights and weights of the 10th standard students of a school.

	Height	Weight
Mean	155 cm	46.50 kg
Variance	72.25 cm^2	28.09 kg

Which is more varying than the other?

Solution For comparing two data, first we have to find their coefficient of variations

Mean $\bar{x}_1 = 155$cm, variance $\sigma_1^2 = 72.25\ cm^2$

Therefore standard deviation $\sigma_1 = 8.5$

$$\text{Coefficient of variation } C.V_1 = \frac{\sigma_1}{\bar{x}_1} \times 100\%$$

$$C.V_1 = \frac{8.5}{155} \times 100\% = 5.48\% \quad \text{(for heights)}$$

Mean $\bar{x}_2 = 46.50$ kg, Variance $\sigma_2^2 = 28.09\ kg^2$

Standard deviation $\sigma_2 = 5.3$ kg

10th_Maths_Chapter 8_English.indd 315 12/16/2024 10:06:39 AM